Introductions

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Thermodynamics (热力学)

- ✓ Thermodynamics: The study of energy and its transformations.
  It is applicable to all fields of science and engineering.
- ✓ Macroscopic thermodynamics: An *empirical science* based on the macroscopic properties of matter and does not rely on any hypotheses concerning the structure of matter.
- ✓ Statistical thermodynamics: It provides *an understanding* of the macroscopic properties of materials, and predicts absolute values based on atomic and molecular structure and motions.

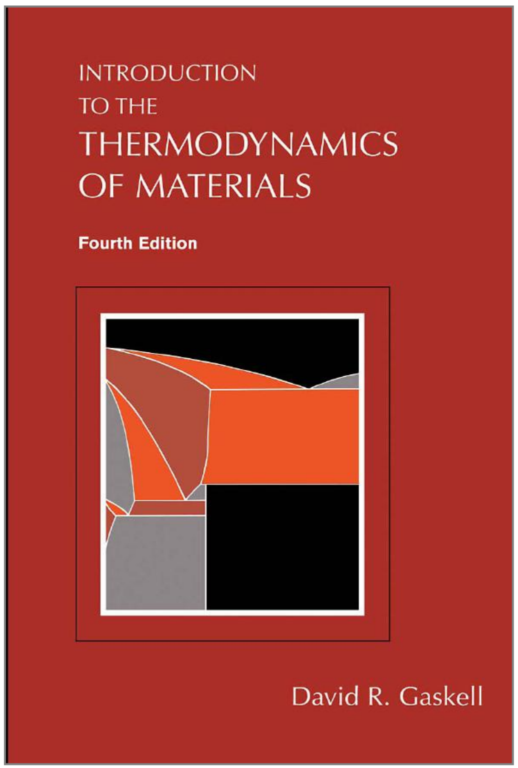
Phase transformations (相变)

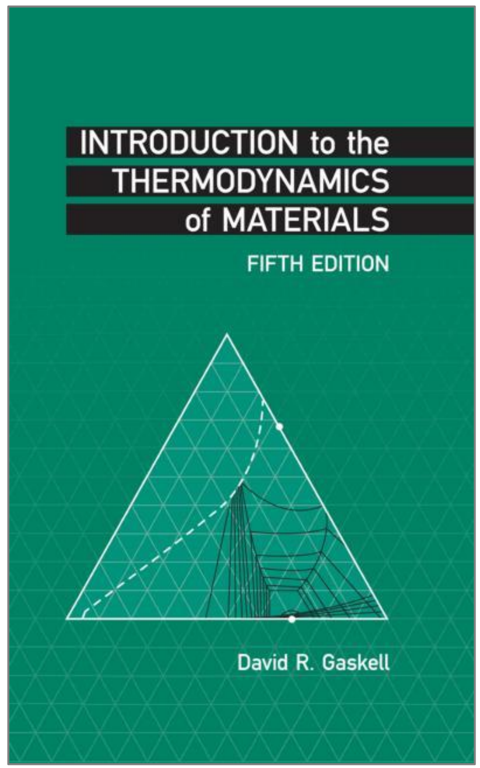
- Phase: A region of space (a thermodynamic system), throughout which *all physical properties of a material are essentially uniform*.
- Phase transformation: The physical processes of transition between the basic states of matter: solid, liquid, and gas, as well as plasma in rare cases.

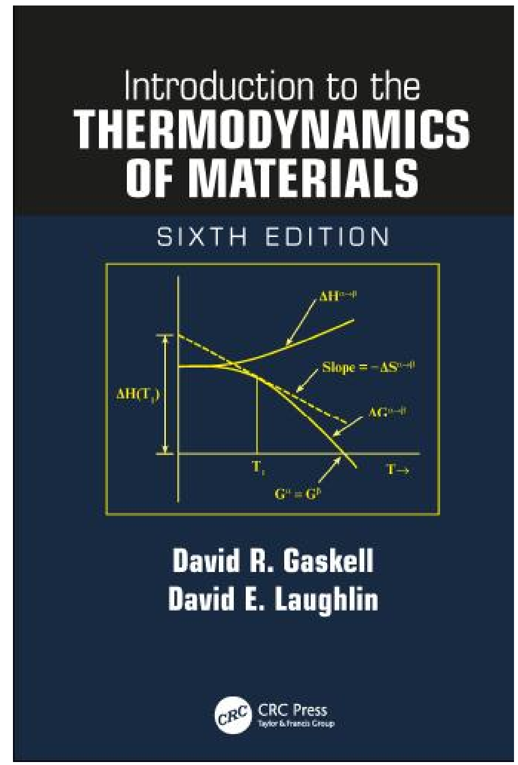
	Solid	Liquid	Gas	Plasma
Solid		Melting	Sublimation	
Liquid	Freezing		Vaporization	
Gas	Deposition	Condensation		Ionization
Plasma			Recombination	

Textbook and references

✓ Purdue University, R. Gaskell "**Introduction to the Thermodynamics of Materials**"
CRC Press, Six Edition





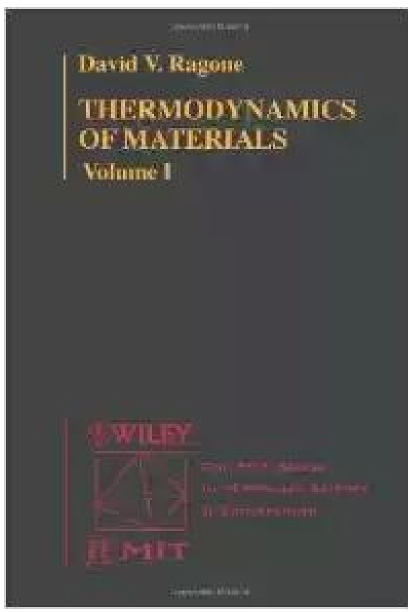


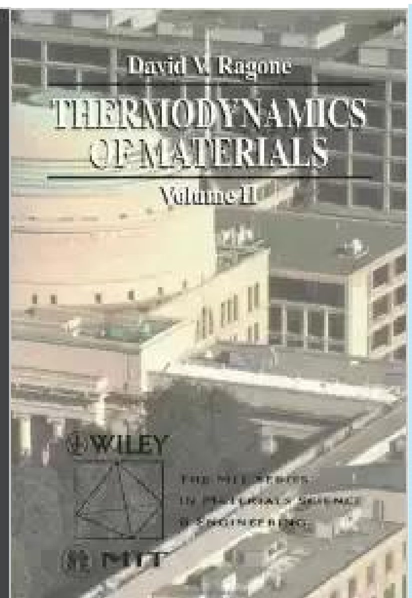
Textbook and references

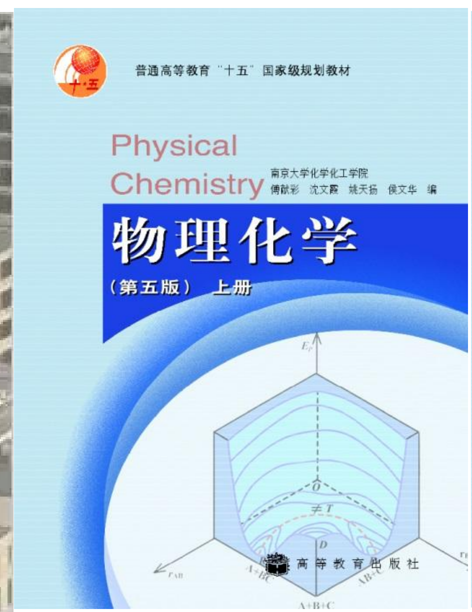
- ✓ MIT David V. Ragone "Thermodynamics of Materials" John Wiley Volume I and Volume II

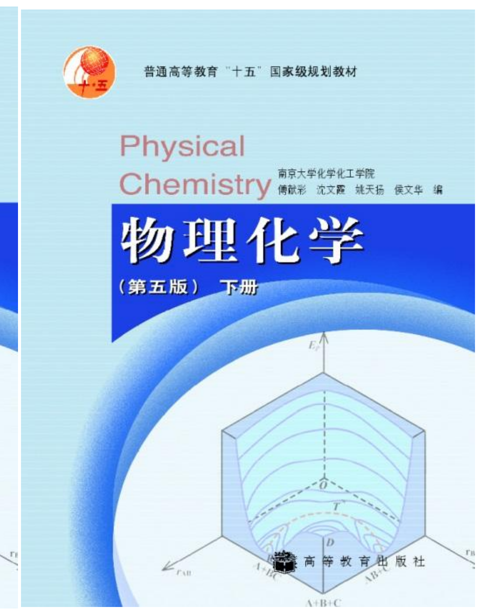
http://ocw.mit.edu/courses/materials-science-and-engineering/3-00-thermodynamics-of-materials-fall-2002/

- ✓ 南京大学 沈文霞等《物理化学》









Schedule

Xiao HAN (韩潇)

Chapter 1 Introduction and definition of terms

Chapter 2 The first law of thermodynamics

Chapter 3 The second law of thermodynamics

Chapter 4 The Statistical Interpretation of Entropy

Chapter 5 Fundamental Equations and their Relationships

Chapter 6 Heat Capacity, Enthalpy, Entropy, and the Third Law of Thermodynamics

Chapter 7 Phase Equilibrium in a One-Component System

Schedule

Xianzong Wang (王显宗)

Chapter 8 The Behavior of Gases

Chapter 9 The Behavior of Solutions

Chapter10 Gibbs Free Energy Composition and Phase Diagrams of Binary Systems

Chapter 11 Reactions Involving Gases

Chapter 12 Reactions Involving Pure Condensed Phases and a Gaseous Phase

Chapter 13 Reaction Equilibria in Systems Containing Components in Condensed Solution

Chapter 14 Electrochemistry

Chapter 15 Thermodynamics of Phase Transformations

Important information

Homework Handing in every Tuesday

Q&A
- During breaks
- QQ group
- Before the exam

Grade
- **30%** normal performance, including homework , attendance, class performance etc.
- **20%** mid-term exam
- **50%** final exam

MOOC link:

https://www.xuetangx.com/course/nwpu0804btintl/19327061?channel=i.area.manual_search

Chapter 1: Introduction and definition of terms

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1.1. Introduction

- ✓ Thermodynamics is concerned with the behavior of matter, where matter is anything that occupies space, and the matter which is the subject of a thermodynamic analysis is called a system.
- ✓ In materials science and engineering, the systems to which thermodynamic principles are applied, are usually chemical reaction systems.
- ✓ The simple thermodynamic system, is the case in which the surroundings interacts with the system only via pressure and temperature changes. The composition remains constant in simple systems.

Systems & Surroundings

- ✓ Thermodynamics is concerned with the behavior of and interactions between portions of the universe denoted as systems and those portions of the universe called the surroundings or the environment.
- ✓ The system is that part of the universe we wish to investigate in detail.
- ✓ The surroundings is that part of the universe outside the system which may interact with it by exchanging energy or matter.

Systems

- ✓ *Isolated systems*: In these systems, no work is done on or by the system. In addition, energy or matter may not enter or leave it.

  The energy of these systems remains constant, as does the overall composition. Isolated systems are therefore unaffected by changes in the surroundings.

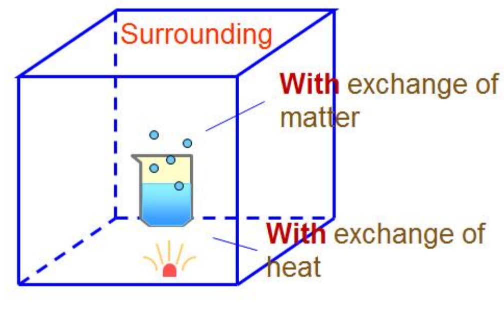
- ✓ *Closed systems*: These are systems which may receive (or give off) energy from (or to) the surroundings.

  The boundaries are called diathermal; that is, they allow thermal energy to transfer through them into or out of the system.

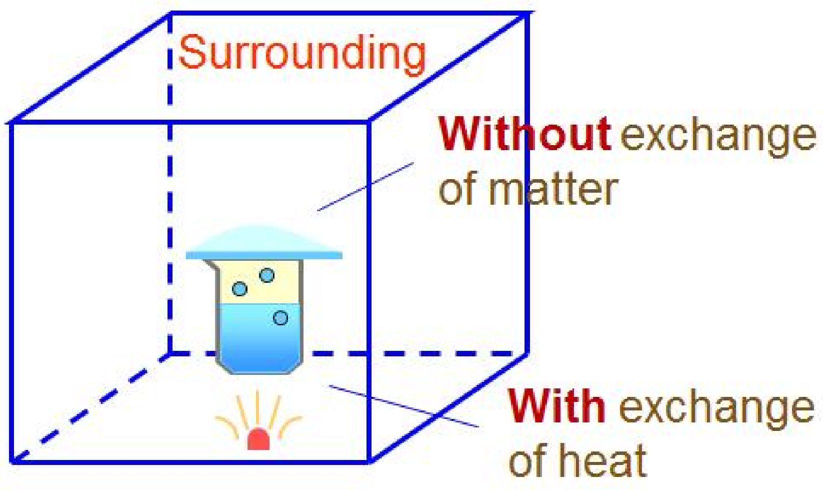
- ✓ *Open systems*: These are systems which can exchange both energy and matter with the surroundings. Neither the energy nor the composition of these systems need remain constant.

Systems

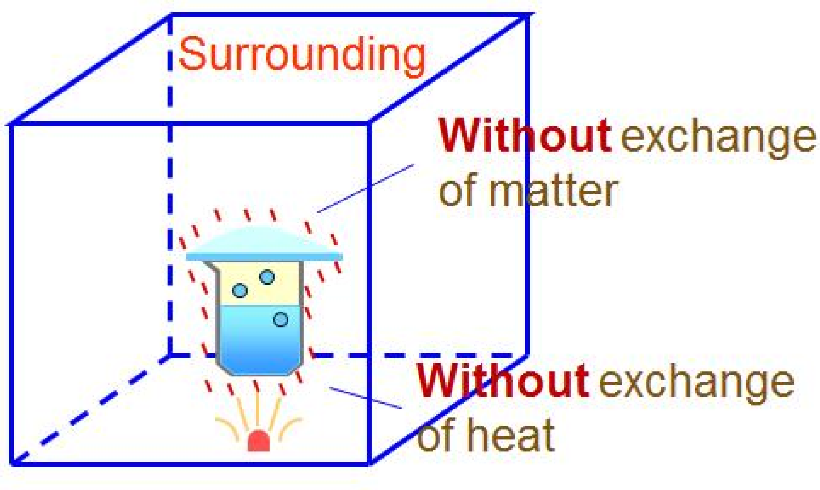
- ✓ Open system (开放体系)
- ✓ Closed system (封闭体系)
- ✓ Isolated system(孤立体系)



Open system

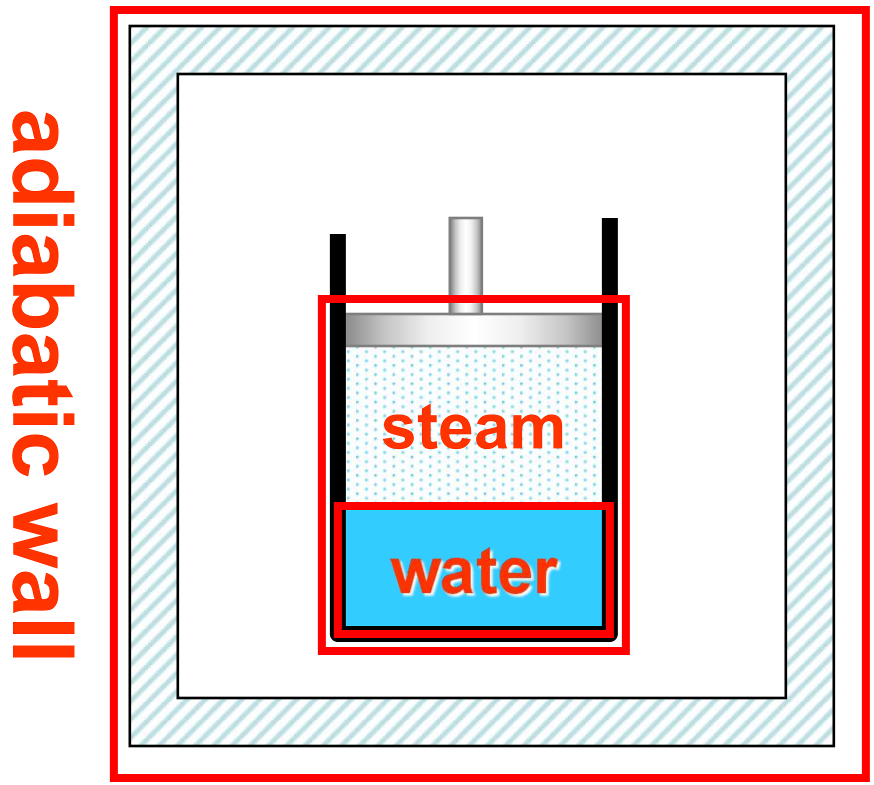


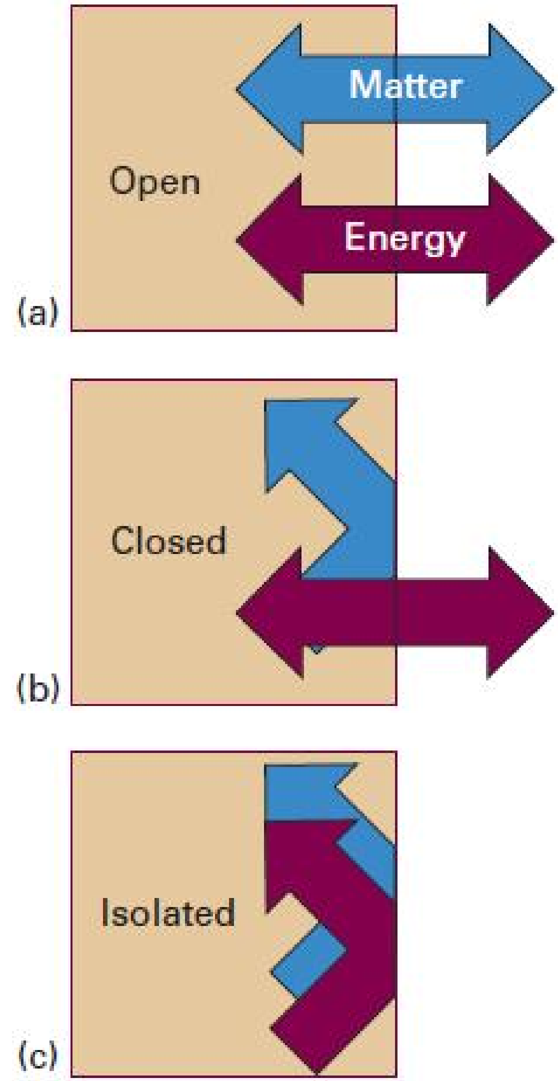
Closed system



Isolated system

Systems





Isolated system

Steam+water is a closed system

Water is a open system

Boundaries

The boundary or wall between the system and the surroundings is what allows such interactions.

- *Adiabatic:* No thermal energy can pass through.
- *Diathermal:* Thermal energy can pass through.
- *Permeable:* Matter can pass through.
- *Impermeable*: Matter cannot pass through.
- *Semipermeable*: Some components are able to pass through, while others are not.

Heat vs. work

✓ Energy transfer between system and surroundings is divided into two categories, ***heat*** and ***work***.

- **Heat** (Q): Energy transferred between the system and surroundings because of a temperature difference. Heat is ***flow quantity*** and is ***energy in transit***.
- **Work** (W): All other forms of energy transferred between the system and surroundings. It can take many forms, e.g. mechanical, electrical, magnetic, etc.

Work

- ✓ Expansion work (volume work) $\delta W = P_e \mathrm{d}V$
- ✓ Non-expansion work

- Mechanical work $\delta W = f_\mathrm{e} \mathrm{d}l$
- Electric work $\delta W = E \mathrm{d}q$
- Surface work $\delta W = \gamma \mathrm{d}A_s$
- ......

Energy of a system

The energy of the system can be divided into three categories:

1. **Internal energy**: It depends on the inherent qualities or properties of the materials in the system, such as composition, physical form and the environmental variables (temperature, pressure, electric field, magnetic field etc.).

- Note1: The mass of the system itself can be considered as a form of energy according to the Einstein's relationship.
- Note2: The form of energy that is related to its temperature is called as thermal energy. ***Heat*** is the energy in transfer between a system and the surroundings. ***Thermal energy*** is possessed by the system.

Energy of a system

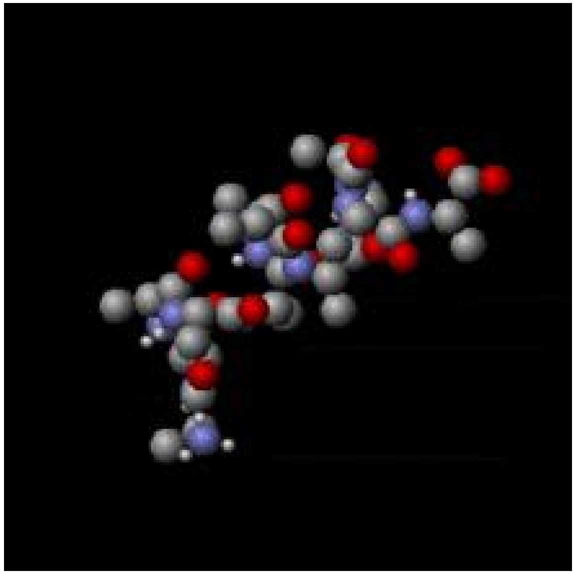
2. Potential energy: It is a sum of the gravitational, centrifugal, electrical and magnetic potential energies.

Isaac Newton (1643-1727) and his apples

Energy of a system

3. **Kinetic energy**: It refers to the energy of the system because of its ***overall*** motion, either translational or rotational. It is the energy of the entire system but not the motion of individual molecules.

YES



NO

Mechanical energy vs. Thermal energy

✓ The discovery that *mechanical energy could be converted to thermal energy* was an important early step in the development of thermodynamics.

Friction (Kinetic energy → Heat)

✓ *The conversion of thermal energy into mechanical work*, or other forms of energy, became a focus of classical thermodynamics.

Heat engines, such as the internal combustion engine used in cars, or the steam engine (Heat → Mechanical energy)

Matter

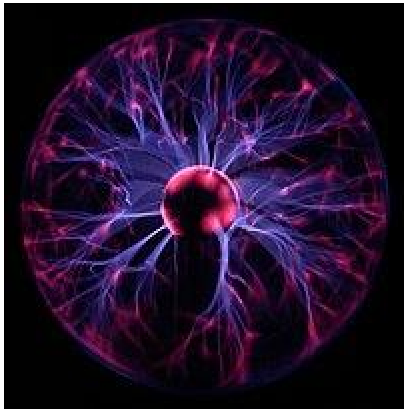
- ✓ Matter is anything that has mass and occupies space.

- ✓ Matter has a given *temperature*, *pressure*, and *chemical composition*, as well as physical properties such as *thermal expansion*, *compressibility*, *heat capacity*, *viscosity*, and so on.

Solid

Liquid

Gas



Plasma

Surroundings

- ✓ A central aim of applied thermodynamics is the determination of the effect of the surroundings on the equilibrium state of a given system.

- ✓ Since the surroundings interacts with the system by transferring or receiving various forms of energy or matter with it, another focus of applied thermodynamics is the establishment of the relationships which exist between the equilibrium state of a given system and the influences which have been brought to bear on it.

1.2. The concept of state

- ✓ *Microscopic state:* the masses, velocities, positions, and all modes of motion of all of the constituent particles in a system. This mass of knowledge determine all of the properties of the system.

- ✓ *Macroscopic state:* the properties of the system. When the values of *a small number of properties* are fixed then the values of all of the rest are fixed

Thermodynamic variables vs. Extrinsic properties

- ✓ The values of the *thermodynamic variables* of a system are not functions of the history of the system; that is, they are independent of the path.
- These thermodynamic variables are *intrinsic* to the state of the system. Such thermodynamic variables are functions of state and can be expressed as *exact differentials* of their dependent variables.

- ✓ The values of the *extrinsic properties* of a system do depend on its history.
- They are *not equilibrium* properties of the system; given time, they may change.

Thermodynamic variables vs. Extrinsic properties

- ✓ An **intrinsic** property is a property of a specified subject that exists itself or within the subject.
- ✓ An **extrinsic** property is not essential or inherent to the subject that is being characterized.

For example, mass is an intrinsic property of any physical object, whereas weight is an extrinsic property that depends on the strength of the gravitational field in which the object is placed.

Example: a simple system

- ✓ When a simple system such as a given quantity of a substance of fixed composition is being considered, the fixing of the values of two of the thermodynamic variables fixes the values of the rest of the thermodynamic variables
- ✓ Only two properties are independent, which, consequently, are called the *independent variables*, and all of the other properties are *dependent variables*.
- ✓ The thermodynamic state of the simple system is thus uniquely fixed when the values of the two independent variables are fixed.

Example: a simple system

- ✓ In the case of the simple system any two properties could be chosen as the independent variables, and the choice is a matter of convenience.

- ✓ Properties most amenable to control are the pressure P and the temperature T of the system.

- ✓ When P and T are fixed, the state of the simple system is fixed, and all of the other properties have unique values corresponding to this state.

Example: change of volume

- ✓ Consider the volume *V* of a fixed quantity of a pure gas as a property, the value of which is dependent on the values of *P* and *T*. The relationship among them are:

$$V = V\left(P, T\right)$$

- ✓ The mathematical relationship of *V* to *P* and *T* for a system is called an *equation of state* for that system, and in a three-dimensional diagram, the coordinates of which are volume, temperature, and pressure, the points in *P*-*V*-*T* space which represent the equilibrium states of existence of the system lie on a surface.

Example: change of volume

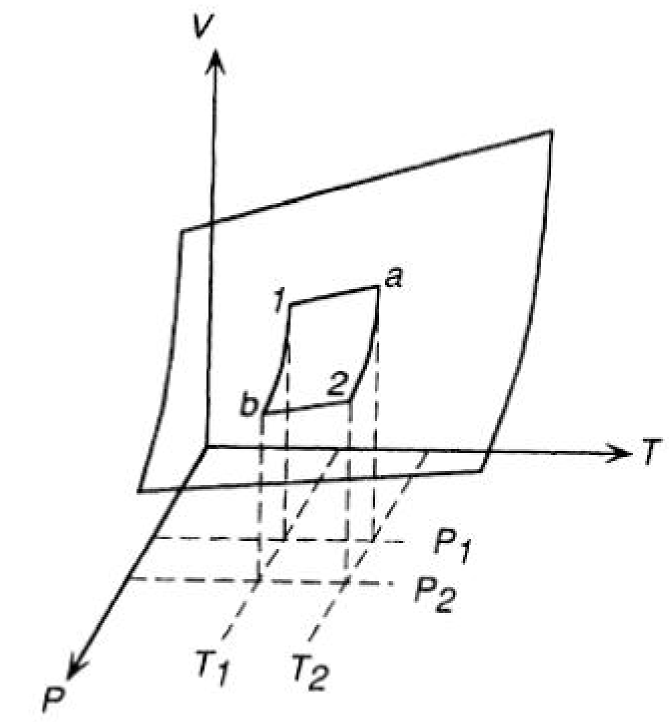


Figure 1.1 The equilibrium states of existence of a fixed quantity of gas in *P*-*V*-*T* space.

✓ Consider a process which moves the gas from state 1 to state 2. This process causes the volume of the gas to change by:

$$\Delta V = V_2 - V_1$$

✓ This process could proceed along an infinite number of paths on the *P*-*V*-*T* surface, two of which, 1→ *a* → 2 and 1 → *b* → 2.

Example: change of volume

- Consider the path 1→*a*→2. The change in volume is:

$$\Delta V = V_a - V_1 + V_2 - V_a$$

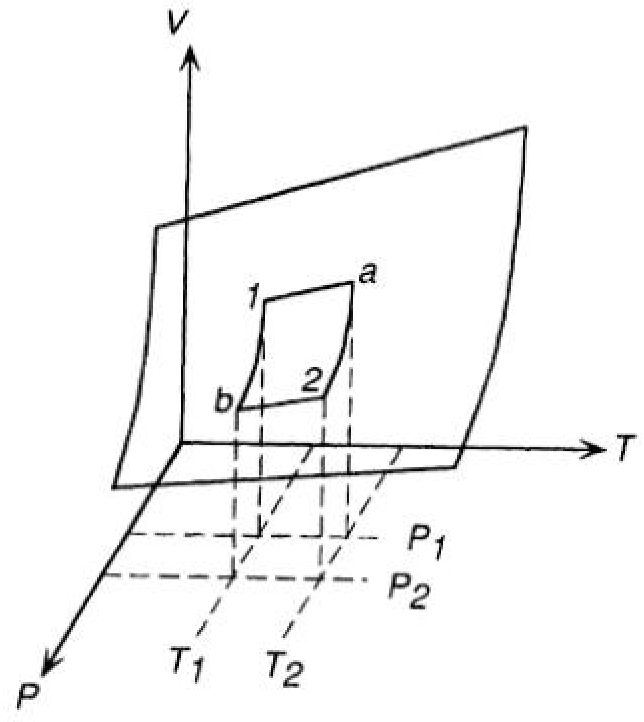


Figure 1.1 The equilibrium states of existence of a fixed quantity of gas in *P-V-T* space.

$$V_a - V_1 = \int_{T_1}^{T_2} \left(\frac{\partial V}{\partial T} \right)_{P_1} dT$$

$$V_2 - V_a = \int_{P_1}^{P_2} \left(\frac{\partial V}{\partial P} \right)_{T_2} dP$$

Example: change of volume

- Consider the path 1→*b*→2. The change in volume is:

$$\Delta V = V_b - V_1 + V_2 - V_b$$

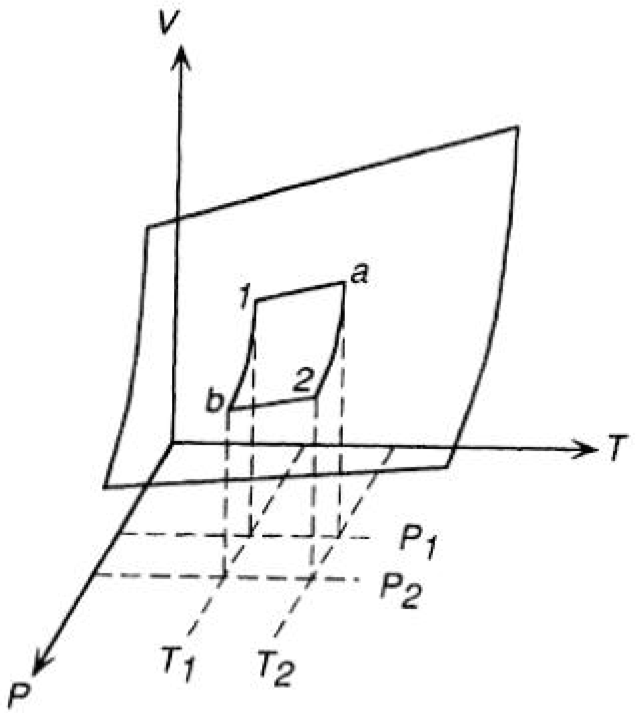


Figure 1.1 The equilibrium states of existence of a fixed quantity of gas in *P-V-T* space.

$$V_b - V_1 = \int_{P_1}^{P_2} \left(\frac{\partial V}{\partial P} \right)_{T_1} dP$$

$$V_2 - V_b = \int_{T_1}^{T_2} \left(\frac{\partial V}{\partial T} \right)_{P_2} dT$$

Example: change of volume

$$\Delta V = \int_{T_1}^{T_2} \left(\frac{\partial V}{\partial T} \right)_{P_1} dT + \int_{P_1}^{P_2} \left(\frac{\partial V}{\partial P} \right)_{T_2} dP$$

$$\Delta V = \int_{P_1}^{P_2} \left(\frac{\partial V}{\partial P} \right)_{T_1} dP + \int_{T_1}^{T_2} \left(\frac{\partial V}{\partial T} \right)_{P_2} dT$$

$$dV = \left(\frac{\partial V}{\partial P} \right)_{T} dP + \left(\frac{\partial V}{\partial T} \right)_{P} dT$$

✓ The change in volume caused by moving the state of the gas from state 1 to state 2 depends only on the volume at state 1 and the volume at state 2 and is independent of the path taken by the gas between the states 1 and 2.

Example: change of volume

- ✓ The isothermal compressibility with dimensions of P^{-1}.

$$\beta_T = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T$$

- ✓ The coefficient of thermal expansion with dimensions of T^{-1}.

$$\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P$$

$$dV = \left(\frac{\partial V}{\partial P}\right)_T dP + \left(\frac{\partial V}{\partial T}\right)_P dT$$

$$\Downarrow$$

$$dV = \alpha V dT - \beta_T V dP$$

1.3. Equilibrium

For a thermodynamic system, one must achieve the following balances at the same time in order to achieve thermodynamic equilibrium:

- ✓ Heat Equilibrium-The temperature of each part of the system is equal; if the system is not adiabatic, the temperature of the system and the surroundings should be equal.
- ✓ Mechanical Equilibrium-The pressure in all parts of the system is equal; there is no relative displacement between the system and the surroundings.
- ✓ Phase Equilibrium-The nature and amount of each phase contained in a system and surroundings do not change over time.
- ✓ Chemical Equilibrium-If the substances in system suffer chemical reaction, the components of the system after equilibrium do not change over time.

Example 1

✓ For any values of temperature and pressure the system is at *equilibrium* only when it has that unique volume which corresponds to the particular values of temperature and pressure.

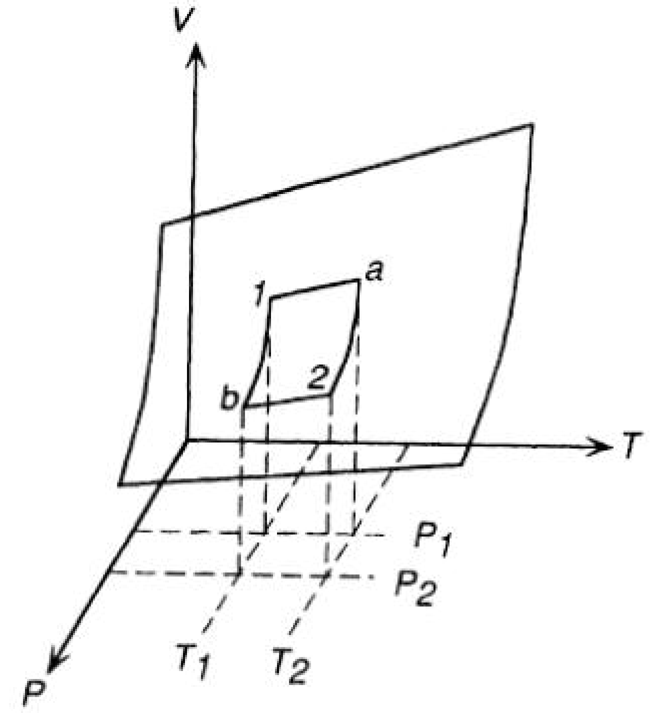


Figure 1.1 The equilibrium states of existence of a fixed quantity of gas in *P-V-T* space.

Example 2

Equilibrium occurs as a result of the establishment of a balance between the tendency of the external influences acting on the system to cause a change in the system and the tendency of the system to resist change.

- The pressure exerted by the gas on the piston equals the pressure exerted by the piston on the gas.
- The temperature of the gas is the same as the temperature of the surroundings (provided that heat can be transported through the wall of the cylinder).

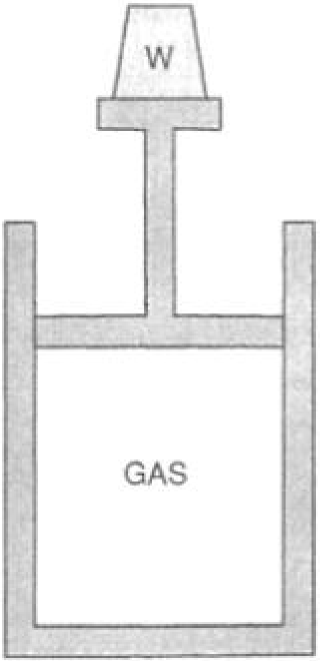


Figure 1.2 A quantity of gas contained in a cylinder by a piston.

1 mole of a gas is shown to be contained in a cylinder by a movable piston.

1.4. The equation of state of an ideal gas

- ✓ The pressure-volume relationship of a gas at constant temperature was determined experimentally in 1660 by Robert Boyle, who found that, at constant *T*.

$$P \propto \frac{1}{V} \quad \text{Boyle's law}$$

- ✓ Similarly, the volume-temperature relationship of a gas at constant pressure was first determined experimentally by Jacques-Alexandre-Cesar Charles in 1787. This relationship, which is known as Charles' law, is, that at constant pressure

$$V \propto T \quad \text{Charles' law}$$

1.4. The equation of state of an ideal gas

Rectangular hyperbolae

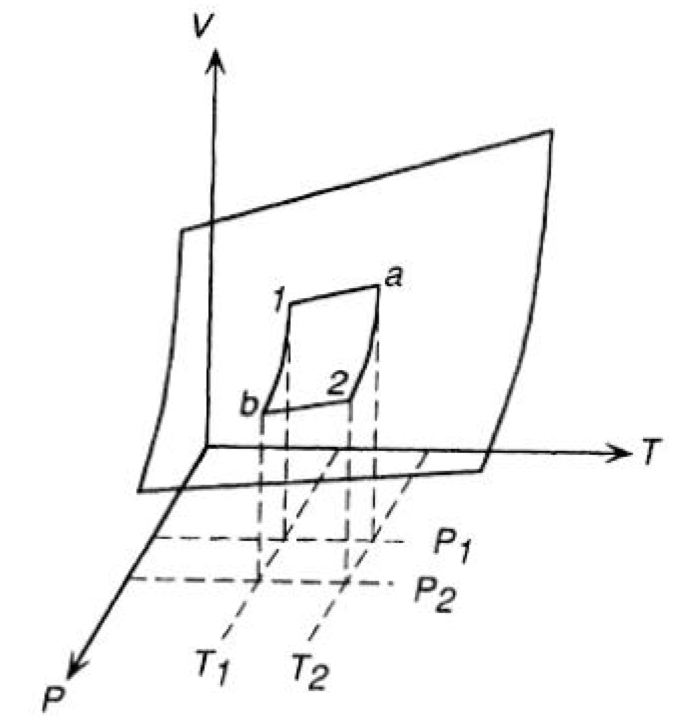


Figure 1.1 The equilibrium states of existence of a fixed quantity of gas in *P-V-T* space.

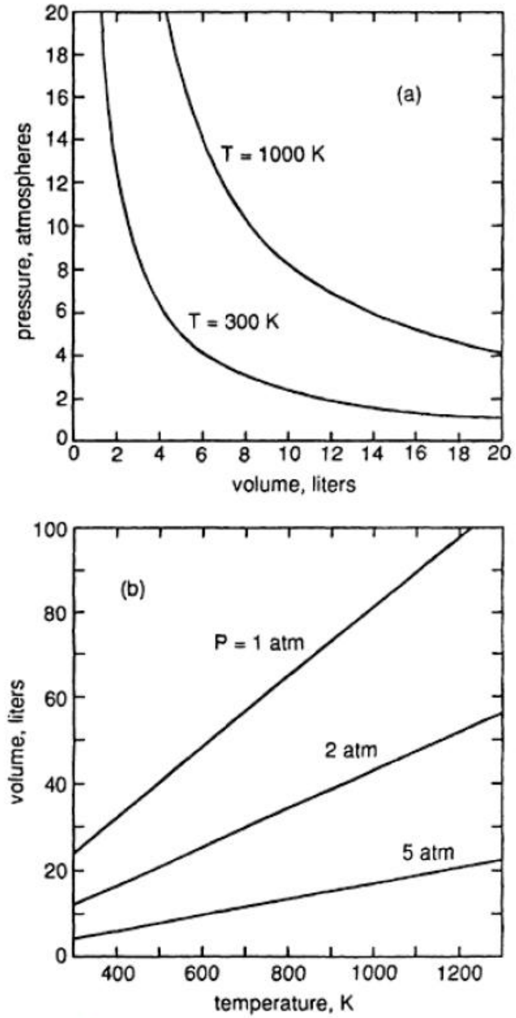


Figure 1.3 *(a)* The variations, with pressure, of the volume of 1 mole of ideal gas at 300 and 1000 K. *(b)* The variations, with temperature, of the volume of 1 mole of ideal gas at 1, 2, and 5 atm.

Straight lines

1.4. The equation of state of an ideal gas

- ✓ In 1802 Joseph-Luis Gay-Lussac observed that the thermal coefficient of what were called "permanent gases" was a constant. The coefficient of thermal expansion, α, is defined as the fractional increase, with temperature at constant pressure, of the volume of a gas at 0 °C; that is.

$$\alpha = \frac{1}{V_0}\left(\frac{\partial V}{\partial T}\right)_P$$

- ✓ Gay-Lussac obtained a value of 1/267 for α, but more refined experimentation by Regnault in 1847 showed α to have the value 1/273.

1.4. The equation of state of an ideal gas

✓ The accuracy with which Boyle's and Charles' laws describe the behavior of different gases varies from one gas to another and that, generally, gases with lower boiling points obey the laws more closely than do gases with higher boiling points.

✓ It was also found that the laws are more closely obeyed by all gases **as the pressure of the gas is decreased**. It was thus found convenient to invent a hypothetical gas which obeys Boyle's and Charles' laws exactly at all temperatures and pressures. This hypothetical gas is called the **ideal gas**, and it has a value of α of 1/273.15.

1.4. The equation of state of an ideal gas

✓ The existence of a finite coefficient of thermal expansion sets a limit on the thermal contraction of the ideal gas; that is, since α=1/273.15 then the fractional decrease in the volume of the gas, per degree decrease in temperature, is 1/273.15 of the volume at 0 °C. Thus, at -273.15 °C the volume of the gas is zero, and hence the limit of temperature decrease, -273.15 °C, is the absolute zero of temperature. This defines an absolute scale of temperature, called the ideal gas temperature scale, which is related to the arbitrary Celsius scale by the equation.

$$T_{\text{degrees absolute}} = T_{\text{degrees celsius}} + 273.15$$

1.4. The equation of state of an ideal gas

Boyle's law $$P_0 V(T, P_0) = PV(T, P)$$

Charles' law $$\frac{V(P_0, T_0)}{T_0} = \frac{V(P_0, T)}{T}$$

P_0=standard pressure (1 atm)

T_0=standard temperature (273.15 degrees absolute)

$V(T,P)$=volume at temperature T and pressure P

$$\frac{PV}{T} = \text{constant}$$

1.4. The equation of state of an ideal gas

✓ From Avogadro's hypothesis the volume per gram-mole of all ideal gases at 0°C and 1 atm pressure (termed standard temperature and pressure—STP) is 22.414 liters.

$$\frac{PV}{T} = \frac{P_0 V_0}{T_0} = \text{constant} = \frac{1\text{atm} \times 22.414\text{liters}}{273.15\text{K} \cdot \text{mole}}$$

$$= 0.082057 \quad \text{liter} \cdot \text{atm/degree} \cdot \text{mole} = R$$ gas constant

$$PV = nRT$$ the *ideal gas law*

1.5. The units of energy and work

- ✓ The unit "liter-atmosphere" occurring in the units of *R* is an energy term. Work is done when a force moves through a distance, and work and energy have the dimensions force×distance. Pressure is force per unit area, and hence work and energy can have the dimensions pressure ×area×distance, or pressure×volume. The unit of energy in S.I. is the joule, which is the work done when a force of 1 newton moves a distance of 1 meter. Liter atmospheres are converted to joules as follows:

$$1 \quad \text{atm} = 101325 \quad \text{newtons/meter}^2$$

$$1 \quad \text{liter} \cdot \text{atm} = 101.325 \quad \text{newton} \cdot \text{meters} = 101.325 \quad \text{joules}$$

$$R = 0.082057 \quad \text{liter} \cdot \text{atm/degree} \cdot \text{mole} = 8.314 \quad \text{joules/degree} \cdot \text{mole}$$

1.6. Extensive and intensive properties

Properties (or state variables) are either extensive or intensive.

- ✓ Extensive properties have values which depend on the size of the system, and the values of intensive properties are independent of the size of the system.

- ✓ Volume is an extensive property, and temperature and pressure are intensive properties. The values of extensive properties, expressed per unit volume or unit mass of the system, have the characteristics of intensive variables; e.g., the volume per unit mass (specific volume) and the volume per mole (the molar volume) are properties whose values are independent of the size of the system.

1.6. Extensive and intensive properties

- ✓ For a system of n moles of an ideal gas, the equation of state:

$$PV' = nRT$$

The volume of the system

- ✓ Per mole of the system, the equation of state is:

$$PV = RT$$

The molar volume of gas, equals to V'/n

1.7. Phase diagrams and thermodynamic components

- ✓ Of the several ways to graphically represent the equilibrium states of existence of a system, the constitution or phase diagram is the most popular and convenient.

- ✓ The complexity of a phase diagram is determined primarily by the number of components which occur in the system, where *components are chemical species of fixed composition*. The simplest components are chemical elements and stoichiometric compounds.

- ✓ Systems are primarily categorized by the number of components which they contain, e.g., one-component (unary) systems, two-component (binary) systems, three-component (ternary) systems, four-component (quaternary) systems, etc.

Phase diagram of water

✓ The phase diagram of a one-component system (i.e., a system of fixed composition) is a two-dimensional representation of the dependence of the equilibrium state of existence of the system on the two independent variables. Temperature and pressure are normally chosen as the two independent variables

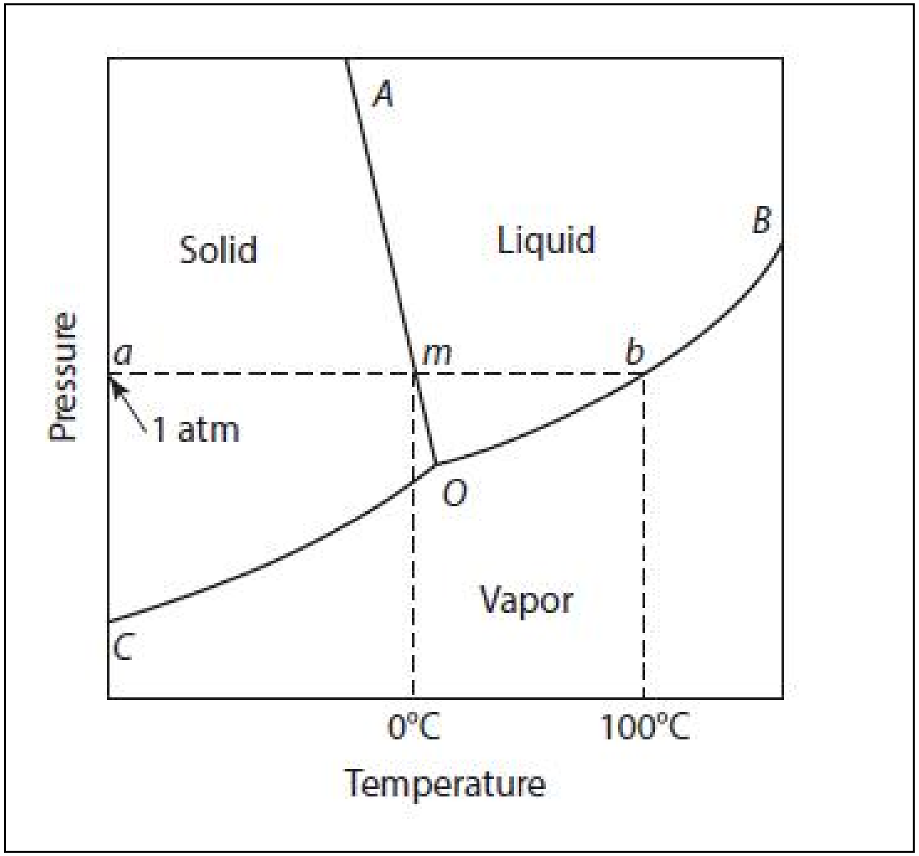


Fig. 1.4 Phase-diagram of water

Phase diagram of water

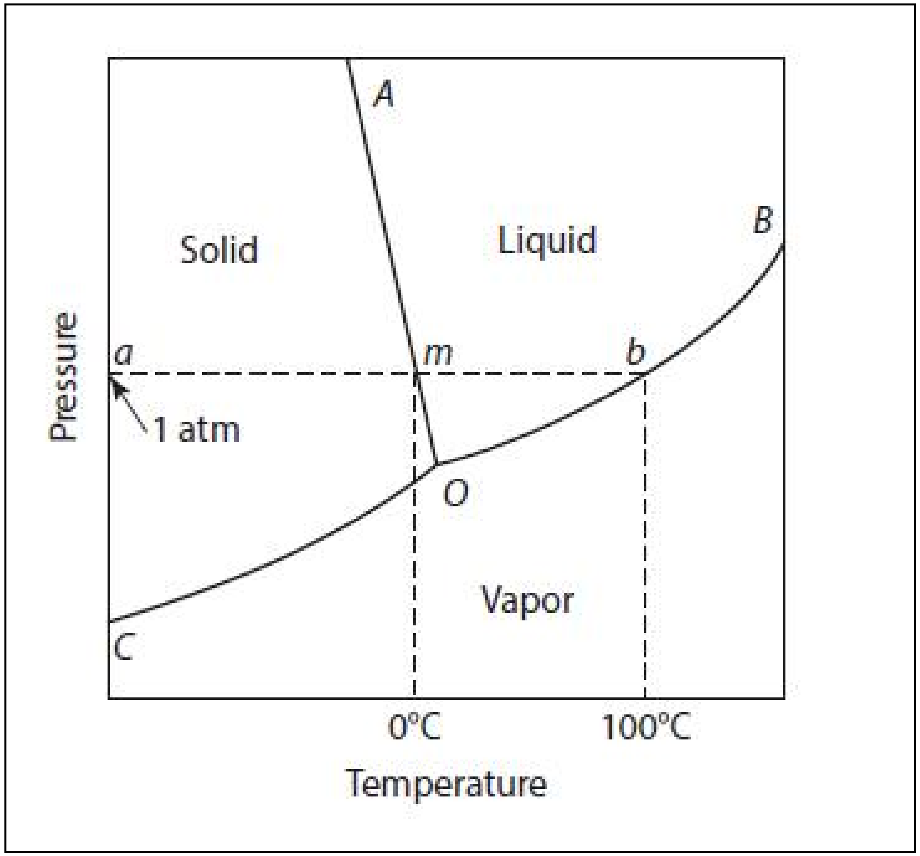
- ✓ A phase is defined as being a finite volume in the physical system with-in which the properties are uniformly constant, i.e., do not experience any abrupt change in passing from one point in the volume to another. Within any of the one phase areas in the phase diagram, the system is said to be homogeneous.

- ✓ The system is heterogeneous when it contains two or more phases, e.g., coexisting ice and liquid water is a heterogeneous system comprising two phases, and the phase boundary between the ice and the liquid water is that very thin region across which the density changes abruptly from the value for homogeneous ice to the higher value for liquid water.

Phase diagram of water

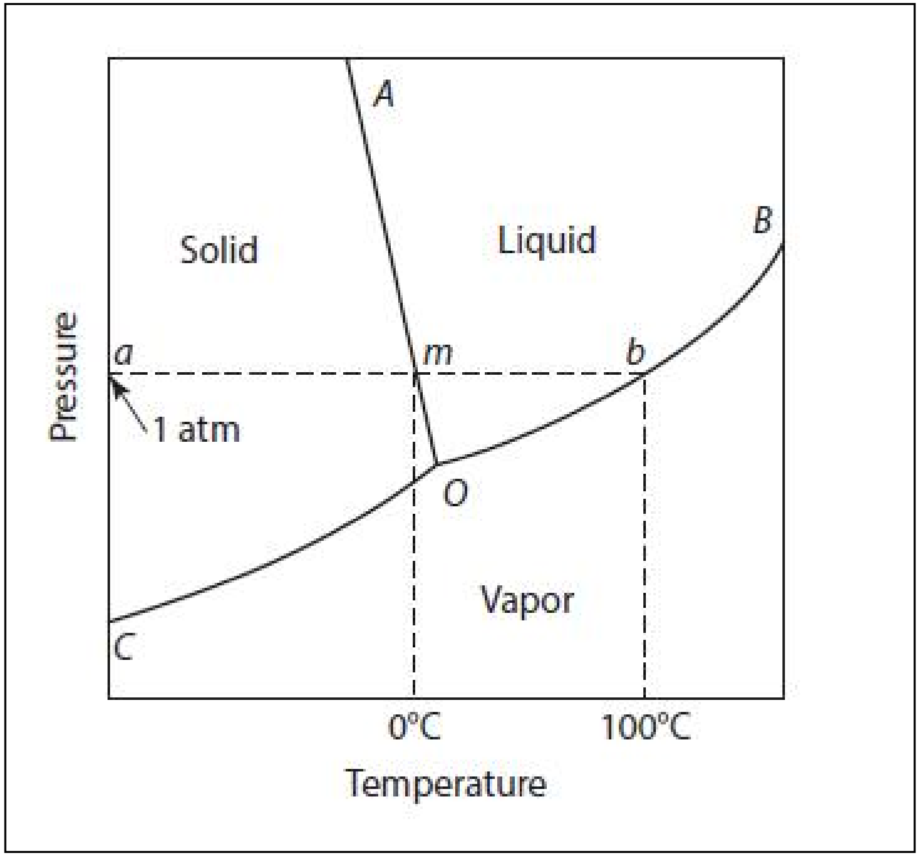
- ✓ The line AO represents the simultaneous variation of *P* and *T* required for maintenance of the equilibrium between solid and liquid H_2O, and represents the influence of pressure on the melting temperature of ice.

- ✓ The lines CO and OB represent the simultaneous variations of *P* and *T* required, respectively, for the maintenance of the equilibrium between solid and vapor H_2O and between liquid and vapor H_2O.



Phase diagram of water

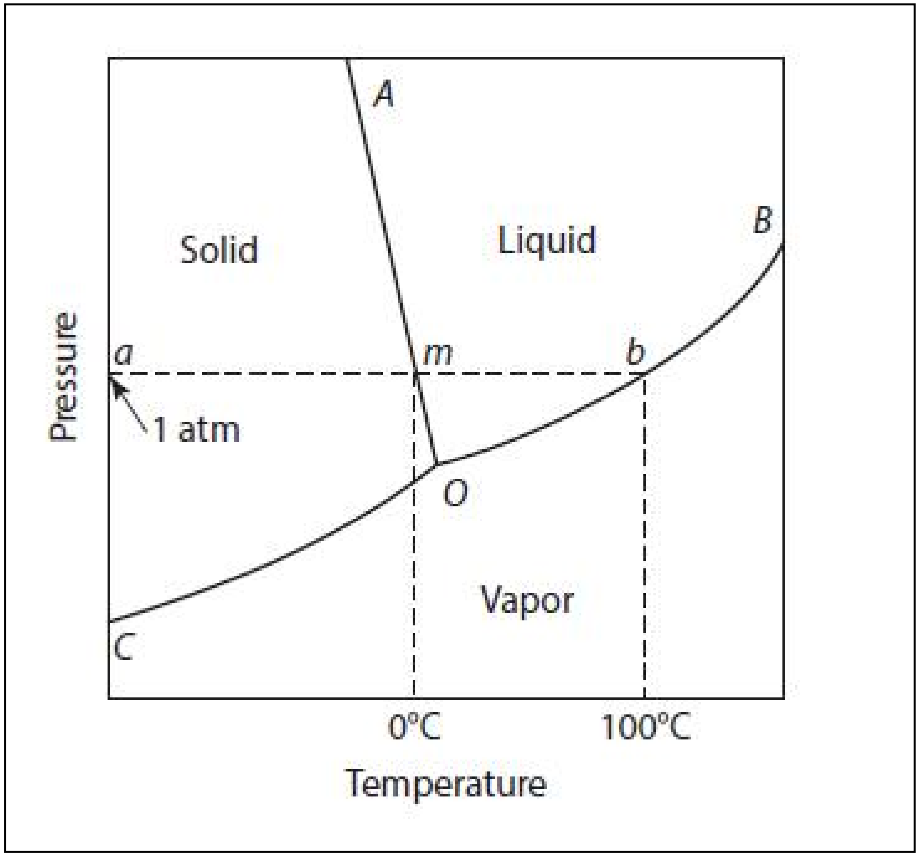
- ✓ The line CO is thus the variation, with temperature, of the saturated vapor pressure of solid ice or, alternatively, the variation, with pressure, of the sublimation temperature of water vapor.
- ✓ The line OB is the variation, with temperature, of the saturated vapor pressure of liquid water, or, alternatively, the variation, with pressure, of the dew point of water vapor.



Phase diagram of water

- ✓ The three two-phase equilibrium lines meet at the point *O* (the triple point) which thus represents the unique values of *P* and *T* required for the establishment of the three-phase (solid+liquid+vapor) equilibrium.

- ✓ The path amb indicates that if a quantity of ice is heated at a constant pressure of 1 atm, melting occurs at the state m (which, by definition, is the normal melting temperature of ice), and boiling occurs at the state b (the normal boiling temperature of water).



Phase diagrams of two-component system

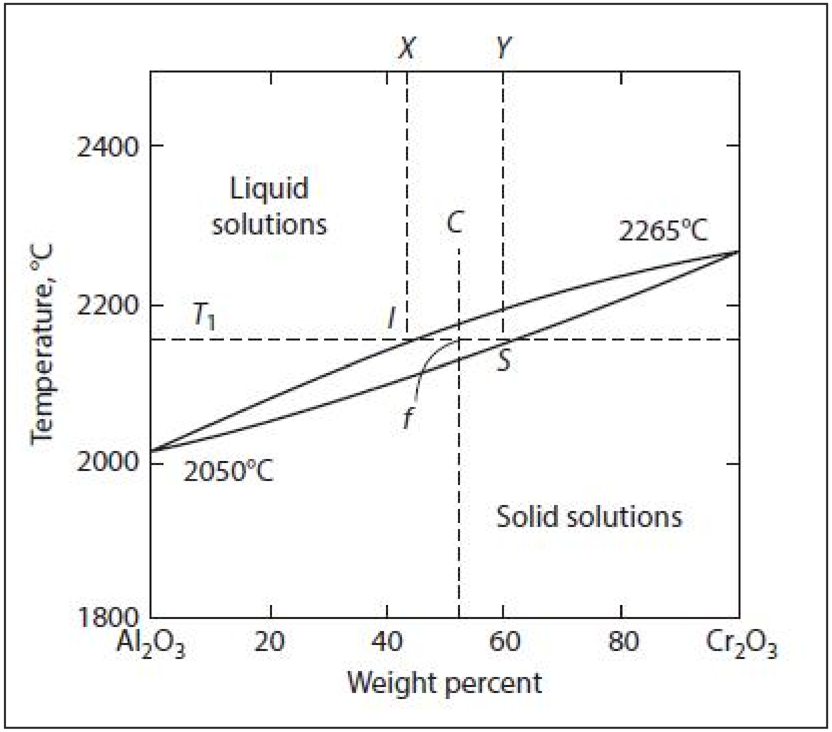
- ✓ Solids may be single phase or may also be composed of more than one phase. It is common to call *a metal composed of more than one component an alloy*. Alloys may be single phase or multiphase. A single phase crystalline alloy consists of two or more components distributed randomly on a single crystal structure. Such single-phase alloys are called *solid solutions* .

- ✓ If the system contains two components, a composition axis must be included and, consequently, the complete diagram is three-dimensional with the coordinates composition, temperature, and pressure.

- ✓ It is sufficient to present a binary phase diagram as a constant pressure section of the three-dimensional diagram. The constant pressure chosen is normally 1 atm, and the coordinates are composition and temperature.

Phase diagrams of two-component system

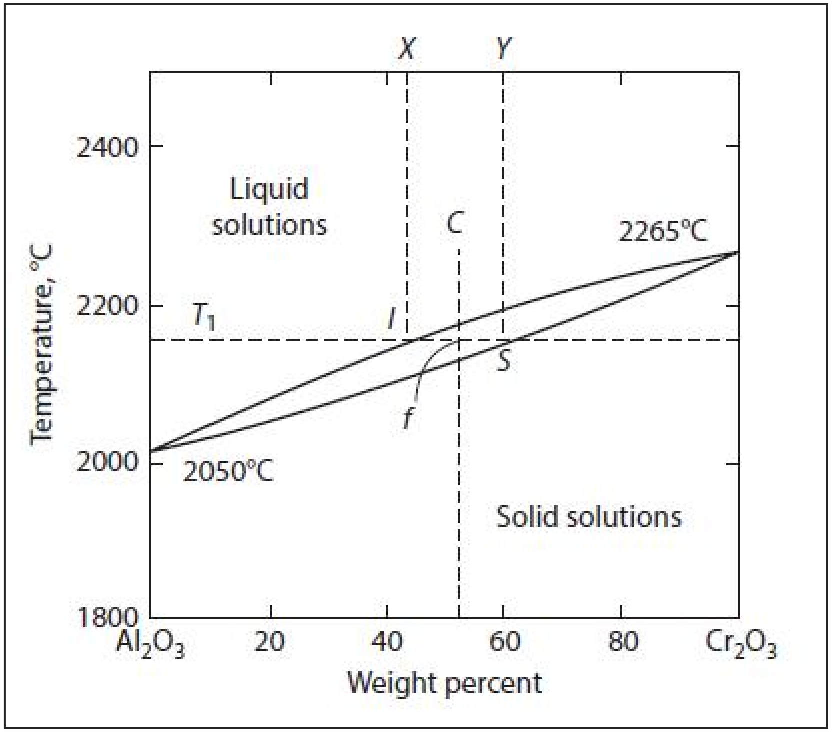
- ✓ At temperatures below the melting temperature of Al_2O_3 (2050 °C), solid Al_2O_3 and solid Cr_2O_3 are completely miscible in all proportions.
- ✓ This occurs because Al_2O_3 and Cr_2O_3 have the same crystal structure and the Al^{3+} and Cr^{3+} ions are of similar size.



- ✓ At temperatures above the melting temperature of Cr_2O_3 (2265 °C), liquid Al_2O_3 and liquid Cr_2O_3 are completely miscible in all proportions.
- ✓ The diagram thus contains areas of complete solid solubility and complete liquid solubility, which are separated from one another by a two-phase area in which solid and liquid solutions coexist in equilibrium with one another.

Phase diagrams of two-component system

- ✓ At the temperature T_1 a Cr_2O_3–Al_2O_3 system of composition between X and Y exists as a two-phase system comprising a liquid solution of composition l in equilibrium with a solid solution of composition s.
- ✓ The relative proportions of the two phases present depend only on the overall composition of the system in the range X–Y and are determined by the lever rule.



- Because the only requirement of a component is that it has a fixed composition, the designation of the components of a system is purely arbitrary. In the system Al_2O_3–Cr_2O_3 the obvious choice of the components is Al_2O_3 and Cr_2O_3.

1.8 Laws of thermodynamics

- ✓ The First Law not only states that the energy of the universe is conserved but it also posits that the various forms of energy (e.g., thermal, electrical, magnetic, and mechanical) can be converted into other forms of energy.
- ✓ When first delineated, it was the fact that thermal energy (heat) could be converted to mechanical work that was of special interest (heat engines). This law also defines an important extensive thermodynamic state function called the internal energy, U, of the system under investigation.

1.8 Laws of thermodynamics

- ✓ Although the Second Law is the one that often gets the most attention in popular discussions of science. This law allows us to make important predictions of the direction in which a system will evolve with time during spontaneous processes, if other important caveats are taken into account.
- ✓ The Second Law introduces another important extensive thermodynamic state function called entropy, *S*. A short version of the Second Law is that the entropy of the universe never decreases.

1.8 Laws of thermodynamics

- ✓ The Third Law states that when a system which is in complete internal equilibrium approaches the absolute zero in temperature, all of the aspects of its entropy approach zero.
- ✓ Sometimes, the Third Law is stated as follows: a system can never be taken to the absolute zero in temperature. This is also called the unattainability principle .

1.8. Laws of thermodynamics

- ✓ The Zeroth Law of Thermodynamics: if system A is in thermal equilibrium with system B, and system B is in thermal equilibrium with system C, then system A is in thermal equilibrium with system C.
- ✓ All three systems must have the same absolute temperature, and therefore, the temperature of a system is a thermodynamic intensive state variable.

Homework

1.11, 1.12 QUALITATIVE EXAMPLE PROBLEMS (P19-P21)

1.2* The expression for the total derivative of V with respect to the dependent variables P and T is

$$dV = \left(\frac{\partial V}{\partial P}\right)_T dP + \left(\frac{\partial V}{\partial T}\right)_P dT$$

Substitute the values of β_T and α obtained in Qualitative Problem 2 into this equation and integrate them to obtain the equation of state for an ideal gas.

1.3* The pressure temperature phase diagram in Figure 1.4 has no two-phase areas (only two-phase curves), but the temperature composition diagram in Figure 1.5 does have two-phase areas. Explain.

The End